

referential words (“I”, etc, which we expected would be disproportionately subjective and thus ones the model would give stereotyped output to), and neither the question nor the answer contained profanity (to avoid refusals). Then we had Llama3-8b-Instruct generate answers to the chosen questions, and the model and human answers were used as “self” and “other” written texts, respectively.

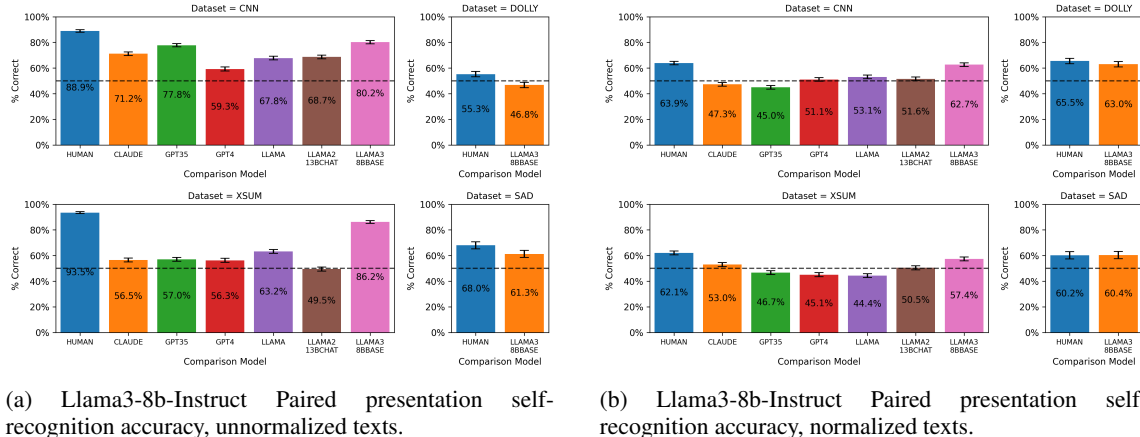
3 RESULTS

3.1 TESTING MODEL SELF-RECOGNITION ABILITY

3.1.1 PAIRED PRESENTATION PARADIGM USING LLAMA3-8B-INSTRUCT

Our first experiment tested whether Llama3-8b-Instruct could achieve above-chance accuracy at self recognition in the Paired paradigm across a range of datasets. As shown in Figure 1a, the model can successfully distinguish its own output from that of humans in all four datasets. In all but the SAD dataset (bottom right panel) it also discriminates its own outputs from that of other AI models. That the model can do so is expected based on prior positive findings, but the level of success is surprisingly high.

In investigating potential causes of this, we noticed that in all datasets except DOLLY, Llama3-8b-Instruct’s output is longer than that of the human output - and frequently than other model output, and that the model used that information (see Figure 8 in Appendix A.6). We include this as an example of how uncontrolled surface differences may allow the model to use information beyond self-knowledge to pass self-recognition tasks. We then took steps to normalize the lengths of all texts, experimenting with filtering down to generated texts that happened to be the same length across authors, and with trimming all texts to a set character length - being careful to trim all texts so as to remove the confound of comparing truncated vs non-truncated texts. Both approaches yielded qualitatively similar results; as the latter afforded us more texts to work with, it is what is used in the results that follow.



(a) Llama3-8b-Instruct Paired presentation self-recognition accuracy, unnormalized texts.

(b) Llama3-8b-Instruct Paired presentation self-recognition accuracy, normalized texts.

Figure 1: Llama3-8b-Instruct Paired presentation self-recognition accuracy with and without length normalization.

Figure 1b shows that length normalization destroys or severely degrades Llama3-8b-Instruct’s ability to distinguish its own generations from that of other AI models (other than the base model’s, which can be degenerate). But it is still able to distinguish its own outputs, albeit at a mostly reduced rate, from human output. This makes sense from the hypothesis of “true” self-recognition, as it is likely that there is commonality in the RLHF process that makes such models’ output relatively easy to distinguish from humans but hard to distinguish from each other’s, and motivates our use of human-written text as the contrast group in our later work. Note that its performance on the DOLLY set actually improves against human output,

presumably because in that dataset human responses were actually longer than its own, and its use of length as a cue was counterproductive.

3.1.2 PERPLEXITY

Another way a model might succeed at the task without actual knowledge of its own writing style is to use the text’s perplexity according to its own representations. But although this would indeed provide a good signal, as Table 3 in Appendix A.1 shows, Llama3-8b-Instruct does not appear to be using it. Correlations between outputs and perplexity were modest (≤ 0.16) and inconsistent. Compared with human output, model output had substantially lower perplexity in all four datasets, but in only one of them was the correlation even marginally statistically significant, and across all comparisons that relationship was usually in the wrong direction. In most cases the model did a worse job of judging its own outputs than it would have if it just relied on perplexity.

3.1.3 PAIRED PRESENTATION PARADIGM USING LLAMA3-8B-BASE

If the chat model is using actual knowledge of its own writing style, presumably that exposure would have come in the course of post-training. Thus, that would imply that the base model would fail at this task. And indeed, that was the case, as shown in Figure 2. (Interestingly, the base model actually did *worse* on the non-length-normalized CNN and XSUM datasets than on the length-normalized ones; subsequent analysis showed that was at least in part because it was using length as a signal, its own outputs were longer, but it was using longer length as a predictor of *other* output. Apparently it was picking up on length as something that differentiated texts and simply guessing what it meant, and happening to guess wrong. This implies that the chat model’s use of length might not have been a confound at all, but rather reflected true knowledge of the kind of outputs it generates.

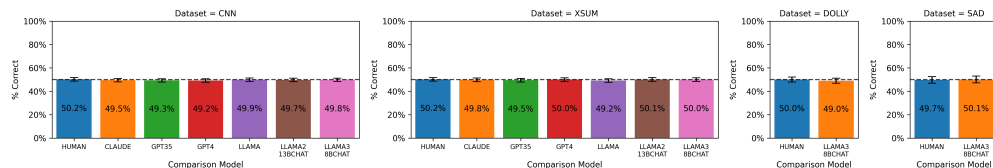


Figure 2: Llama3-8b-base Paired presentation self-recognition accuracy, normalized texts.

3.1.4 INDIVIDUAL PRESENTATION PARADIGM

A more challenging task, and one which would remove some of the confounds that come in a paired presentation, is to show the model a single piece of text and ask it whether it wrote it. This is what we do in the Individual presentation paradigm. In this paradigm, a model could still in theory use its experience with human idioms to make a “familiar/unfamiliar” judgment; if this were all that was happening, one would expect the base model to do as well on the task as the chat model. As shown in Table 1, the chat model is successful at differentiating its own output from that of humans in three of the four datasets. The base model, however, is not able to distinguish texts it has generated from texts generated by others. And once again, perplexity was not correlated with model judgments.

3.2 A “SELF-RECOGNITION” VECTOR

3.2.1 ISOLATING A VECTOR

We used contrasting pairs of self- and other-written texts, followed by out-projection of nuisance components (see Methods) to isolate directions in the residual stream that corresponded to the choice of claiming self or other authorship. The final token before the model gave its response was identified as the most promising target. Figure 7 in Appendix A.5 shows the Tuned Lens decoding of it: the positive direction contains a number of tokens related to self-recognition (e.g., “my”, “I”, “match”), while the negative direction contains a number of tokens pointing in the other direction (e.g., “their”, “different”, “other”).

Table 1: Llama3-8b-Instruct Individual presentation self-recognition accuracy.

Dataset	Chat model		Base model	
	%Correct	P-Value	%Correct	P-Value
CNN	57.2	0.004	50.0	1.000
XSUM	61.0	0.0001	50.3	0.9045
DOLLY	52.3	0.3576	49.5	0.8415
SAD	65.8	0.0001	48.5	0.5485

We tested the relevance of this vector for model output by steering (Turner et al., 2024) with it (adding scaled versions of it to activations in the residual stream during generation), at a range of multipliers and layers. As can be seen in Figure 3, steering with multipliers in the 3 to 6 range on layers 14-16 was most effective, achieving 100% effectiveness at causing the model to claim authorship, regardless of true authorship, or, when steered in the opposite direction, deny authorship, for both self- and human-written texts.

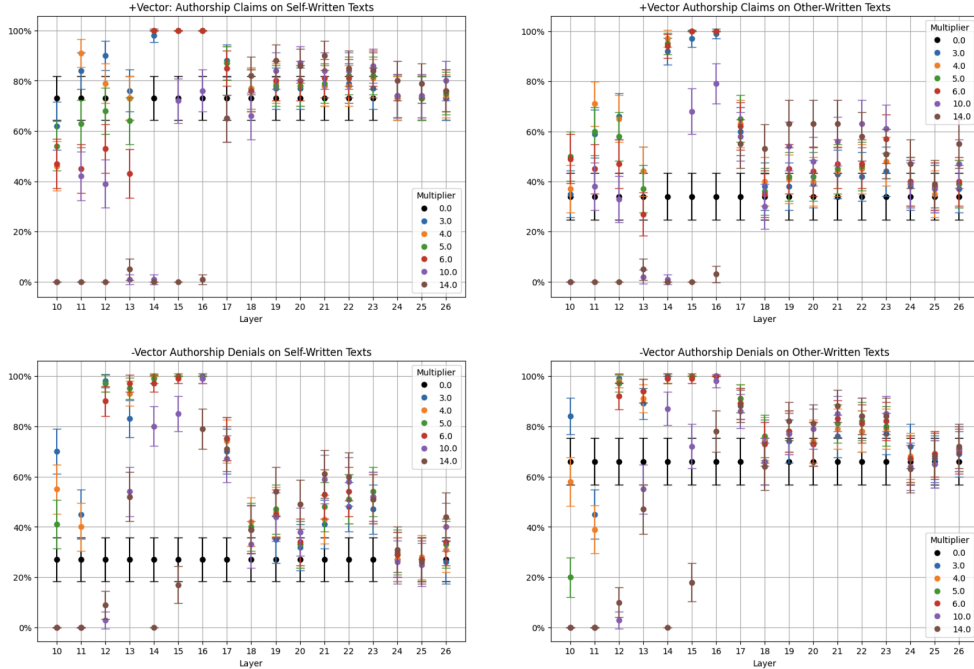


Figure 3: Steering effectiveness by layer and multiplier for Individual presentation paradigm test set 1. +/- Vector: positive/negative steering. Black dots are unsteered model. Colors indicate multipliers; for example, as can be seen in the upper right, positive steering with multiplier 10 and layer 16 led the model to claim authorship of a text that it did not write $\sim 80\%$ of the time, as compared with the unsteered model’s $\sim 35\%$.

This pattern was very consistent. Figure 4 combines positive and negative steering for self and other written texts to create a map of aggregate effectiveness for two new datasets, one comprised of test-set texts from the Individual paradigm, and one from the QA dataset. Early layers show little or negative steering effectiveness (high multipliers can lead to degenerate output in the early layers especially), then there’s a sudden increase in effectiveness at moderate multipliers at layer 14 that peaks at layer 16, and then quickly drops off.